Mira Chami

Applied AI Engineer

London, UK · +44 7495 468016 · miranchami@gmail.com · mirachami.dev

Applied AI engineer building grounded generation, knowledge-constrained dialogue and controlled model actions inside interactive products. Experience spans game engines, backend services, cross-platform desktop applications and shipped games.

PROFESSIONAL EXPERIENCE

Tencent

Nov 2023 – Present

Game Tech R&D Engineer · London, UK

Investigative Game & Applied AI

Gemini, Node.js, Godot

- Owned Godot case-file, evidence-board and interview UI: source-linked text selection, drag-to-pin cards, connections/annotations, board navigation, animated transitions and keyboard/pointer controls.
- Built a **multi-stage Gemini/Google Search-grounded research pipeline** for source gathering, evidence extraction, witness research and narrative generation, with JSON Schemas, cross-reference checks, retrieved-source citations, versioned case packages and a comparison viewer.
- Implemented knowledge-constrained AI witnesses distinguishing facts, beliefs and player hypotheses, with **verified replies**, one revision attempt and source-attributed verbatim clues. Players review questions from up to three selected evidence cards and an intent before sending.
- Built Node.js/Express services with request validation, session restoration, caching, timeouts, cancellation and retries; automated schema/data-integrity and UI tests across viewport sizes with mocked replies, without live model calls.

LLM Game Director & Designer Tools

Godot, GDScript

- Built server-authoritative co-op multiplayer with Steam P2P, LAN, listen/dedicated servers, replication, RPCs, sessions and local multi-client testing; implemented data-driven runs and UI for objectives, death/revival, node/map selection, currency, rewards and win/loss.
- Built the LLM director’s validated action library for spawning enemies, granting item rewards and changing progression, with shared gameplay events and entity context as model input.
- Built a timeline runtime and Godot editor plugin that designers used to create behaviour timelines for the AI director without code: shared actions, typed parameters, time/event-driven triggers, pacing previews and validation before model integration.
- Profiled procedural generation and low frame rates with Godot’s profiler, identifying CPU-bound bottlenecks.

Desktop AI Companion

React, Rust, Tauri

- Owned React/JavaScript and Rust client architecture; evaluated Electron/Tauri prototypes for performance, bundle size, multi-window behaviour and native APIs, then migrated to Tauri v2.
- Built prompt submission with session/game context, HTTP/SSE streaming, progressive replies and persistent per-user history; chat, companion, avatar and settings windows with shared messaging, overlays, hotkeys, tray access, gRPC, Three.js and PixiJS.
- Configured Windows/macOS CI setup, builds, installers and packaging; added structured logging and diagnostic panels, with Steam integration optional through feature flags.

Online Services & Partner Integration

UE5, C++, Blueprints

- Implemented replicated UE5 gameplay and UI in C++/Blueprints: Free-for-All and Teams matches, character state, team rules, scoring, interactables and match transitions.
- Integrated InGame, Tencent’s public online services framework, covering authentication, lobbies, matchmaking, quests, progression, inventory and achievements; supported partner studios with gameplay, UI, plugin and backend integration.
- Enabled PlayStation 5 support and local dedicated-server debugging in Unreal Editor; configured and modified CI pipelines.

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PROFESSIONAL EXPERIENCE (CONTINUED)

The Video Games Studio (VGS)

Jun 2020 – Aug 2022

Game Programmer · Beirut, Lebanon

- **Shipped a Unity mobile runner** covering gameplay, difficulty tuning, monetisation, PlayFab integration and release; built a separate tool to manage competition cycles and retrieve winner data.
- Owned gameplay architecture for a published Unity puzzle game: Adventure/Drawing modes, shared progression and saves, and saving/replaying player-created pixel-art levels.
- Led programming for a Unity adventure prototype, translating the GDD into character, ability and interaction systems; built touch controls with the Unity Input System alongside level designers.
- Built configurable difficulty, progression and shop content with ScriptableObjects; implemented Unity UI and integrated PlayFab, third-party plugins and backend services across projects.
- Supported technical interviews, onboarding and mentoring, code reviews and Jira/Trello task planning; contributed to GDDs and maintained gameplay-system and software-architecture documentation.

TECHNICAL TOOLKIT

Applied AI:
LLM Integration · Context Management · LLM Tool Design · Google Search Grounding · Structured Outputs (JSON Schema) · LLM Output Validation · LLM Response Streaming · Source Attribution

Languages:
Python · TypeScript · JavaScript · C++ · C# · Rust · GDScript · Java · C

Backend & Networking:
Node.js · Express · HTTP · SSE · gRPC · Protobuf · InGame · PlayFab · Steamworks · Photon PUN 2

Engines & Desktop:
Unreal Engine 5 · Unity · Godot · Blueprints · React · Tauri · Electron

Build & Engineering:
Git · Perforce · CI · Automated Testing · Mocking · Build & Release Automation · Structured Logging · Performance Profiling

Graphics & Native APIs:
Three.js · PixiJS · Unity URP · Native API Integration

EDUCATION

MSc Games Development (Programming) · Kingston University London · 2022 – 2023

BSc Computer Science · American University of Beirut · 2017 – 2021

BA Economics · American University of Beirut · 2016 – 2020

MISCELLANEOUS

Languages: English, French and Arabic - native

Additional projects: Portal rendering/DualSense triggers · 3D map UI · Photon/PlayFab arena · Sudoku solver